Candidates Seat No:________________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M.Tech. Theory Exam (CL)

May 2018

CL 768 Design of Composite Structures

Date: 08.05.2018, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

(Use of IS: 800-2007,IS-11384-1984, IS-456,Steel tables and Euro code 4 are permitted)

SECTION - I

Q - 1 (a) Write Any Two. **[10]**

1) Explain with neat sketch interaction curve for the uni- axial bending of column.
2) Explain using any case studies "Composite structure is better than R.C.C or Steel Structure".
3) List the main functions of shear connectors? What are the different types of shear connectors?

Q - 2 (a) The composite column of size 450 mmX450 mm and 3 m in length is pinned ended. The column is under the design axial load of 1100 kN and bending moment about XX axis is of 140 kNm.It is encased with steel section ISHB 350@72.4 kg/m is at the center. Steel reinforcement is 4 No's of 14 mm dia. bars. **[20]**

Check the adequacy of the Section for uniaxial bending. Adopt Grade of Concrete M30 and Grade of Steel Fe 415.

Assume the position of neutral axis in the web.

Zpax= 1131.6 $*10^3$ mm^3 and Zpay=199.4$*10^3$ mm^3

$$h_n = \frac{A_c p_{ck} - A_s'(2p_{sk} - p_{ck})}{2b_c p_{ck} + 2t_w(2p_y - p_{ck})}$$

Take moment resistance ratios

$$\mu_x = \frac{(\chi_x - \chi_d)}{(1-\chi_c)\chi_x} \qquad \text{when } \chi_d \geq \chi_c$$

$$= 1 - \frac{(1-\chi_x)\chi_d}{(1-\chi_c)\chi_x} \qquad \text{when } \chi_d < \chi_c$$

OR

Q - 2 (a) (i) Explain the design concepts for the checking the adequacy of concrete encased Composite section for biaxial bending. **[15]**

(ii) Explain the seismic behavior of composite structures. **[05]**

Q - 3 (a) How to calculate the plastic resistance of concrete filled tubular composite column? **[05]**

OR

Q - 3 (a) What is second order effect? Explain Criteria to consider second order effect. **[05]**

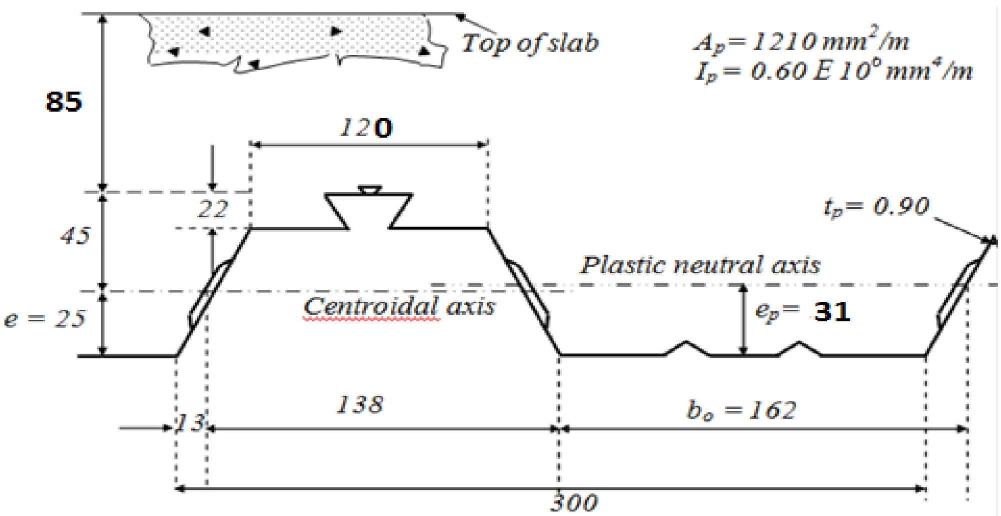
SECTION - II

Q - 4(a) Check the adequacy of the continuous composite profiled deck slab with spans of 3.3 m. The cross section of the profiled sheeting is shown in Fig. The Slab is propped at the centre during construction stage. [Υ_{ap} = 1.15, γ_{vs} = 1.15 , M20 concrete is used], Live load=4.2 kN/m^2, dead load= 2.1 kN/m^2, Construction load=1.5 kN/m^2 **[20]**
Yield strength of steel, f_{yp} = 280 N/mm^2

Design thickness, allowing for zinc, t_p = 0.9 mm, Effective area of cross-section, A_p = 1210 mm^2/m, Moment of inertia, I_p = 0.6 * 10^6 mm^4/m. Plastic moment of resistance M_{pa} = 5 kN-m/m, Distance of centroid above base e = 25 mm , Distance of plastic neutral axis above base, e_p = 31 mm ,Resistance to vertical shear, V_{pa} = 49.2 kN/m. For resistance to longitudinal shear, m = 184 N/mm^2, k = 0.0530 N/mm^2 Modulus of elasticity of steel, E_a = 2.0 * 10^5 N/mm^2 , Width of flange of beam, bf = 150 mm

Design vertical shear, Vd = 28.3 kN/m , Basic shear strength, τ_{Rd} = 0.3 N/mm^2

Resistance to vertical shear, $V_{v,Rd}$, $V_{v,Rd} = (b_0/b)d_p \ , \ \tau_{Rd}\, k_v(1.2 + 40\rho)$



Q - 5 (a) Design a simply supported composite beam with 9 m span. The thickness of slab is 120mm. The floor is to carry an imposed load of 2.5kN/m^2, partition load of 1.5kN/m^2, construction load of 0.5kN/m^2 and floor finish load of 0.5kN/m^2. The distance between two beams is 3 m. Take M30 grade concrete and Fe415 steel and **[10]**

density of concrete is $24kN/m^3$. Take ISMB 450 section as a beam, $E = 2*10^5\ N/mm^2$.

(1) Check for the moment capacity for the construction stage and composite stage.
(2) Check for lateral buckling of the top flange. Design of shear connector.

OR

Q - 5(a) Design a simply supported composite beam with 9.1m span. The thickness of slab is 125mm. The floor is to carry an imposed load of $2.5kN/m^2$, partition load of $1.5kN/m^2$, construction load of $0.5kN/m^2$ and floor finish load of $0.5kN/m^2$. The distance between two beams is 3.2m. Take M30 grade concrete and Fe415 steel and density of concrete is $24kN/m^3$. Take ISMB 450 section as a beam, $E = 2*10^5\ N/mm^2$. **[10]**

(1) Check for all serviceability criteria.
(2) Check for transverse reinforcement.

Q - 6 (a) Write a role of profiled metal decking. Explain serviceability requirements for design of composite slabs **[05]**

OR

Q - 6 (a) Write a note on serviceability requirements for design of composite beams. **[05]**

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